YASH DONGA

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Summary

Final-year B.Tech student specializing in Artificial Intelligence and Machine Learning at CHARUSAT with hands-on experience across full-stack development, backend engineering, and applied AI/agentic systems. Proven ability to design and deploy end-to-end products spanning PostgreSQL-backed data pipelines, semantic search with vector embeddings, and local LLM integration, alongside strong machine learning fundamentals in model training, evaluation, and REST API deployment.

Education

Charotar University of Science and Technology (CHARUSAT) **2023 – 2027**
B.Tech in Artificial Intelligence & Machine Learning | Minors: Data Science & ML

The Imperia Science School – Dhoraji **2022 – 2023**
Higher Secondary Certificate (HSC) – Science Stream

Technical Skills

Programming Languages: Python, C++, SQL

Web & Backend Development: Flask, Streamlit, REST APIs, HTML, CSS

Databases & Data: PostgreSQL, pgvector, MySQL

AI & Machine Learning: Supervised/Unsupervised Learning, Neural Networks (ANN, CNN), NLP, Multi-Agent AI, Semantic Search, Embeddings, Model Training & Evaluation, Hyperparameter Tuning

Libraries & Frameworks: Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Plotly, OpenCV

Tools & Platforms: GitHub, Vercel, Ollama, Jupyter Notebook, Google Colab, Kaggle, Tableau, Power BI

Experience

Data Science Intern – Cognifyz Technologies **May 2026 – Jun 2026**
Restaurant Data Analysis | EDA, Feature Engineering, Predictive Modeling

- Performed multi-level exploratory data analysis on a dataset of 9,000+ restaurant records, uncovering key business insights through descriptive statistics, geospatial analysis, and trend visualization.
- Engineered new features including name length, address length, and binary-encoded categorical variables; conducted correlation analysis to identify high-impact predictors for model training.
- Trained and evaluated Linear Regression, Decision Tree, and Random Forest models for aggregate rating prediction; Random Forest achieved **$R^2 = 0.986$** and RMSE = 0.177, outperforming baseline by over 40%.

Machine Learning Intern – Exposys Data Labs **May 2025 – Jun 2025**
Profit Prediction System | Random Forest Regressor, Flask, Model Deployment

- Designed and deployed a Profit Prediction System using a Random Forest Regressor model served via a Flask REST API, enabling real-time predictions through a web interface.
- Executed a complete ML pipeline including data cleaning, feature encoding, normalization, cross-validation, and production deployment, reducing prediction error through iterative hyperparameter tuning.

Projects

Wayfind – AI-Powered Retail Navigation System | *PostgreSQL, pgvector, MiniLM Embeddings* **2026**

- Built a full-stack in-store navigation platform where customers scan a QR code, search for products, and receive walking directions, including multi-floor routes with stairs.
- Engineered an AI-driven CSV import pipeline that cleans messy product data, groups similar products using semantic embeddings, auto-generates store sections, and validates safe walking paths.
- Implemented semantic search using a local open-source embedding model (MiniLM) with pgvector, enabling meaning-based matching (e.g., "sport shoes" correctly retrieving "Nike running footwear") without paid APIs.
- Validated with 150 real-world messy product records and 135 automated tests; deployed to production on Vercel.

AI-Powered Analytics & BI Dashboard Platform | *Python, Flask, Pandas, Plotly, Ollama* **Jun 2026**

- Developed a full-stack business intelligence platform that ingests raw datasets and generates interactive dashboards using a multi-agent AI architecture with specialized agents for data science, visualization, backend, QA, and orchestration.
- Automated an end-to-end pipeline covering data profiling, anomaly detection, and AI-driven insight generation via a local LLM (Ollama); engineered a Power BI-inspired UI with KPI cards, drill-down analytics, and CSV/PDF export, reducing manual reporting time significantly.

AI-Powered Travel Planning Agent | *Python, Streamlit, Gemini API, Google Maps API* **Jan 2026**

- Built an intelligent travel itinerary generator integrating the Gemini LLM, Google Maps API, and Weather API to produce real-time, personalized day-by-day travel plans, complete with a hotel recommendation engine, automated packing list generation, and exportable PDF itineraries via Streamlit.

Certifications

Data Structures and Algorithms in Java – NPTEL | Data Analysis using Python – Samatrix | Machine Learning with Python – IBM (Coursera)